

AHSEN NAZIR

AI Software Engineer

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PERSONAL STATEMENT

AI Engineer with 5+ years of experience building and deploying production grade AI systems across Computer Vision, NLP, and Generative AI. Specialized in LLM systems, RAG pipelines, multi-agent workflows, workflows and scalable AI automation in applied and enterprise environments. Experienced in international deployments across China, US, and Southeast Asia.

Dev interests: *NLP, GenAI, Computer Vision*

CORE TECHNICAL SKILLS

Artificial Intelligence & ML: Computer Vision, NLP, Generative AI, LLMs, Multimodal AI, Model Fine-tuning

LLM Systems Engineering: Retrieval-Augmented Generation (RAG), Prompt Engineering & Optimization, AI Evaluation (Evals), Inference optimization with LoRA/qLoRA, Autonomous Agents, Vector Databases (FAISS / Milvus/Chroma / etc.), Context Window Management, LLM Deployment & Monitoring, Multilingual LLM Expertise: Urdu, Arabic, English, Russian

Frameworks & Libraries: PyTorch, HuggingFace Transformers, Detectron2, Stable Diffusion, LangChain

Backend & APIs: FastAPI, REST APIs, Celery, RabbitMQ, Redis, Ray

MLOps & DevOps: Docker, Docker Compose, Linux, Multi-GPU Training, Weights & Biases

Databases & Cloud: PostgreSQL, SQL, AWS, GCP, Azure

Languages: Python, C++, SQL

PROFESSIONAL EXPERIENCE

National Centre for Physics

Islamabad, Pakistan

Assistant Manager (Artificial Intelligence), Deep Learning Lab

April 2023 – Apr 2026

- Led production AI systems for document intelligence and LLM automation across 3 integrated systems in an applied research environment.
- Designed and deployed RAG pipelines with embeddings and vector databases enabling semantic search over 10K+ documents, improving retrieval accuracy and reducing latency.
- Built multi-agent systems using n8n orchestrating 20+ step workflows for end-to-end automation, reducing manual effort by 30–50%.
- Developed tool-using autonomous agents integrating 5+ external APIs for real-time decision-making and execution across dynamic workflows.
- Implemented LLM evaluation and API orchestration systems with rate limiting, streaming, and reliability controls achieving 99%+ production stability.

Shenzhen AiMall Technology Co. Ltd.

Shenzhen, China

Intern

Dec 2024 – February 2025

- Developed Stable Diffusion-based image animation pipelines using pose and audio inputs across 3+ configurations.
- Optimized ComfyUI workflows (FaceAlignment, MediaPipe, InsightFace), improving generation efficiency and output stability.
- Integrated vision and audio models for multimodal animation systems enabling synchronized generation across 2 modalities with real-time inference support.
- Reduced memory usage by 50% via model offloading and compute optimization enabling large diffusion workflows on constrained GPUs.
- Implemented multi-GPU DiT-style parallelization across 2 GPUs (T4, 1080Ti), improving throughput.

Freelance (Upwork)

Remote

AI Developer

Jun 2020 – Jun 2025

- Deployed deep learning models on custom datasets supporting 1000+ concurrent inference requests under production workloads.
- Built secure, rate-limited AI APIs with authentication and controlled model exposure.
- Developed distributed AI systems using multi-threading and parallel execution for low-latency inference.
- Designed 50+ reusable business logic modules across modular AI services enabling scalable system architecture.
- Delivered end-to-end AI solutions for clients across USA, Japan, Malaysia, with production deployment and monitoring.

EDUCATION

Pakistan Institute of Engineering and Applied Sciences (PIEAS)

Masters of Science in Computer Science

* CGPA: 3.65/4.00

Islamabad, Pakistan

Feb 2021 – Nov 2022

Bahria University Islamabad Campus (BUIC)

Bachelor of Science in Computer Science

Islamabad, Pakistan

Sep 2015 – Feb 2020

PROJECTS

Portfolio

WORKSHOPS AND TRAININGS

49th International Nathiagali Summer College (INSC)

2024

– Organized by Pakistan Atomic Energy Commission (PAEC), focusing on AI and Allied Disciplines

Workshop on Cyber Weapons and Malware Analysis

2024

– Institute of Space Technology (IST)

Young Officers Orientation Course

2023

– Pakistan Centre of Excellence for Nuclear Security (PCENS)

2-Month AI Fellowship

2022-2023

– Organized by Artificial Intelligence Technology Centre. Secured **1st** Position in the batch

Organized 4x National Artificial Intelligence Seminar (NAIS)

2022-2025

– CoE AITeC, National Centre for Physics (NCP)

ACADEMIC HONORS, PRIZES, AND FELLOWSHIPS

IT Endowment Scholarship

2021

– Awarded a fully funded scholarship for M.S. studies at Pakistan Institute of Engineering and Applied Sciences (PIEAS), sponsored by the institute.

AI Fellowship – National Centre for Physics

2023

– Ranked 1st in cohort. Selected for a 2-month Artificial Intelligence fellowship leading to a job offer at the National Centre for Physics.

Poster Competition Winner – ISAIMP 2023

2023

– Secured 2nd Prize in the poster competition at the International Symposium on Artificial Intelligence, Medical Imaging, and Pattern Recognition (ISAIMP-2023).

– Poster Title: “Precision Agriculture: Wheat Crop Classification and Yield Estimation”.